SATGURU STUDY CENTRE
Chennai - 81

TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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SECOND MID TERM TEST - 2024

MATHEMATICS

Time Allowed : 1.30 Hours] **[Max. Marks : 50**

PART - I

I. **Choose the correct Answer.** 7x1=7

1. Which of the following is a solution of the equation 2x-y=6

(a) (2,4) (b) (4,2) (c) (3,-1) (d) (0,6)

2. The value of K for which the pair of Linear equations 4x+6y-1=0 and 2x+ky-7=0 represent parallel lines is

(a) k=3 (b) k=2 (c) k=4 (d) k=-3

3. The interior angle made by the side in a parallelogram is 90° then the parallelogram is -------

(a) Rhombus (b) Rectangle

(c) Trapezium (d) Kite

4. The points whose ordinate is 4 and which lies on the y axis is --------

(a) (4,0) (b) (0,4) (c) (1,4) (d) (4,2)

5. If (x+2, 4) = (5, y-2), then the coordinates (x,y) are -----

(a) (7, 12) (b) (6, 3) (c) (3, 6) (d) (2, 1)

6. If $P\left(\frac{a}{3}, \frac{b}{2}\right)$ is the midpoint of the line segment joining A (-4,3) and B (-2,4) then (a,b) is --------

(a) (-9,7) (b) (-3,7/2) (c) (9,-7) (d) (3,-7/2)

7. If (1, -2), (3,6), (x,10) and (3,2) are the vertices of the parallelogram taken in order then the value of x is ----------

(a) 6 (b) 3 (c) 4 (d) 5

PART - II

II. **Answer any 5 of the following Questions. Q.No. 14 is compulsory.** 5x2=10

8. Solve by the method of elimination. 2x-y=3; 3x+y=7

9. The sum of the numerator and denominator of a fraction is 12. If the denominator is increased by3, the fraction becomes $\frac{1}{2}$. Find the Fraction

10. The angles of a quadrilateral are in the ratio 2 : 4 5 7 Find all the angles

11. Find the distance between the points (-4, 3) and (2, -3)

12. The centre of a circle is (-4,2). If one end of the diameter of the circle is (-3,7), then find the other end

13. Find the Centroid of the triangle whose vertices one A(6,-1), B(8,3) and C(10, -5)

14. The length of the diagonals of a Rhombus are 12cm and 16cm. find the side of the Rhombus.

PART - III

III. Answer any 5 of the following questions. (Q.No.21 is compulsory) 5x5=25

15. The middle digit of a number between 100 and 1000 is zero and the sum of the other digit is 13 If the digits are reversed, the number so formed exceeds the Original number by 495.Find the numbers.

16. Solve: 3x - 4y=10 and 4x + 3y = 5 by the method of Cross multiplication.

17. In a parallelogram ABCD, the bisectors of the consecutive angles ∠A an ∠B intersect at P show that ∠APB = 90°

18. Show that the points A(7,10), B(-2,5) and C(3,-4) are the vertices of a right and angled triangle.

19. The MidPoint (x,y) of the line joining (3,4) and (p,7) lies on 2x+2y+1=0, then what will be the value of P.

20. What are the coordinates of B if Point P(-2,3) divides the line segment joining A(-3,5) and B internally in the ratio 1:6?

21. Find the length of median through A of a triangle vertices are A (-1,3), B (1,-1) and C (5,1)

PART - IV

IV. Answer the following questions. 1x8=8

22. (a) Solve Graphically: x+y=7; x-y=3 **(OR)**

(b) Plot the following Points in the coordinate Plane. What type of Geometrical shape is formed.

(-3, 3), (2,3), (-6, -1) and (5, -1)